

BIOLOGY SIMS SIUT

CELL STRUCTURE & FUNCTION - 20 MCQs

1. The control center of the cell is:

- A) Mitochondria
- B) Nucleus ✓
- C) Ribosome
- D) Golgi apparatus

Answer: B

2. Which organelle is known as the powerhouse of the cell?

- A) Nucleus
- B) Ribosome
- C) Mitochondria ✓
- D) Lysosome

Answer: C

3. The site of protein synthesis in the cell is:

- A) Nucleus
- B) Ribosome ✓
- C) Golgi apparatus
- D) Endoplasmic reticulum

Answer: B

4. Which organelle is responsible for photosynthesis?

- A) Mitochondria

- B) Lysosome
- C) Chloroplast ✓
- D) Ribosome

Answer: C

5. The fluid inside the cell is called:

- A) Cytoplasm ✓
- B) Nucleoplasm
- C) Cell sap
- D) Plasma

Answer: A

6. Which organelle is involved in the packaging and secretion of proteins?

- A) Endoplasmic reticulum
- B) Ribosome
- C) Lysosome
- D) Golgi apparatus ✓

Answer: D

7. The cell wall of plant cell is made up of:

- A) Cellulose ✓
- B) Chitin
- C) Protein
- D) Lipid

Answer: A

8. Which organelle contains digestive enzymes?

- A) Ribosome
- B) Lysosome ✓
- C) Mitochondria

D) Nucleus

Answer: B

9. The largest organelle in the cell is:

- A) Mitochondria
- B) Golgi apparatus
- C) Nucleus ✓
- D) Endoplasmic reticulum

Answer: C

10. Which organelle is involved in lipid synthesis?

- A) Rough ER
- B) Smooth ER ✓
- C) Ribosome
- D) Lysosome

Answer: B

11. The semi-permeable membrane surrounding the cell is called:

- A) Cell wall
- B) Cytoplasm
- C) Plasma membrane ✓
- D) Nuclear membrane

Answer: C

12. Ribosomes are composed of:

- A) DNA and protein
- B) RNA and protein ✓
- C) Lipid and protein
- D) Carbohydrate and protein

Answer: B

13. Which organelle is absent in prokaryotic cells?

- A) Ribosome
- B) Cell membrane
- C) Nucleus ✓
- D) Cell wall

Answer: C

14. The inner folded membrane inside mitochondria is called:

- A) Cristae ✓
- B) Stroma
- C) Thylakoid
- D) Matrix

Answer: A

15. Which organelle is responsible for detoxification of drugs?

- A) Rough ER
- B) Smooth ER ✓
- C) Lysosome
- D) Golgi apparatus

Answer: B

16. The fluid-filled space inside chloroplast is called:

- A) Cristae
- B) Stroma ✓
- C) Matrix
- D) Cytoplasm

Answer: B

17. Which of the following is a membrane-bound organelle?

- A) Ribosome
- B) Nucleus ✓
- C) Centriole
- D) Nucleolus

Answer: B

18. The organelle involved in cell division in animal cells is:

- A) Centriole ✓
- B) Chloroplast
- C) Lysosome
- D) Ribosome

Answer: A

19. The nuclear envelope is composed of:

- A) Single membrane
- B) Double membrane ✓
- C) Triple membrane
- D) No membrane

Answer: B

20. Which organelle has 70S ribosomes in it?

- A) Nucleus
- B) Mitochondria ✓
- C) Golgi apparatus
- D) ER

Answer: B

BIOLOGICAL MOLECULES - 20 MCQs

1. The most abundant molecule in living organisms is:

- A) Protein
- B) Lipid
- C) Water ✓
- D) Carbohydrate

Answer: C

2. The building blocks of proteins are:

- A) Monosaccharides
- B) Nucleotides
- C) Amino acids ✓
- D) Fatty acids

Answer: C

3. Which of the following is a monosaccharide?

- A) Sucrose
- B) Maltose
- C) Glucose ✓
- D) Lactose

Answer: C

4. The bond between amino acids is called:

- A) Glycosidic bond
- B) Phosphodiester bond

- C) Peptide bond ✓
- D) Hydrogen bond

Answer: C

5. Enzymes are made up of:

- A) Carbohydrates
- B) Lipids
- C) Proteins ✓
- D) Nucleic acids

Answer: C

6. The building blocks of nucleic acids are:

- A) Amino acids
- B) Nucleotides ✓
- C) Monosaccharides
- D) Fatty acids

Answer: B

7. Which type of lipid is the main component of cell membrane?

- A) Triglyceride
- B) Cholesterol
- C) Phospholipid ✓
- D) Steroid

Answer: C

8. Starch is an example of:

- A) Monosaccharide
- B) Disaccharide
- C) Polysaccharide ✓
- D) Oligosaccharide

Answer: C

9. The genetic material in most organisms is:

- A) DNA ✓
- B) RNA
- C) Protein
- D) Carbohydrate

Answer: A

10. Glucose and fructose are:

- A) Isomers ✓
- B) Isotopes
- C) Polymers
- D) Same compound

Answer: A

11. The sugar found in DNA is:

- A) Ribose
- B) Deoxyribose ✓
- C) Glucose
- D) Sucrose

Answer: B

12. Which of the following is a disaccharide?

- A) Glucose
- B) Fructose
- C) Sucrose ✓
- D) Starch

Answer: C

13. The process of converting excess glucose to glycogen is called:

- A) Glycolysis
- B) Glycogenesis ✓
- C) Glycogenolysis
- D) Gluconeogenesis

Answer: B

14. Which type of RNA carries genetic information from DNA to ribosomes?

- A) rRNA
- B) tRNA
- C) mRNA ✓
- D) All of these

Answer: C

15. The function of tRNA is to:

- A) Carry genetic code
- B) Carry amino acids ✓
- C) Form ribosomes
- D) Replicate DNA

Answer: B

16. Which of the following is a saturated fatty acid?

- A) Oleic acid
- B) Linoleic acid
- C) Palmitic acid ✓
- D) Arachidonic acid

Answer: C

17. The enzymatic activity of ribozymes is associated with:

- A) Protein
- B) DNA
- C) RNA ✓
- D) Lipid

Answer: C

18. The most abundant protein in human body is:

- A) Hemoglobin
- B) Myoglobin
- C) Collagen ✓
- D) Keratin

Answer: C

19. Which test is used to detect the presence of reducing sugars?

- A) Biuret test
- B) Benedict's test ✓
- C) Iodine test
- D) Xanthoproteic test

Answer: B

20. The structural formula of cholesterol contains:

- A) Four fused rings ✓
- B) Two fused rings
- C) Three fused rings
- D) Five fused rings

Answer: A

ENZYMES - 20 MCQs

1. Enzymes are biological:

- A) Carbohydrates
- B) Catalysts ✓
- C) Lipids
- D) Nucleic acids

Answer: B

2. The protein part of an enzyme is called:

- A) Coenzyme
- B) Prosthetic group
- C) Apoenzyme ✓
- D) Holoenzyme

Answer: C

3. The non-protein organic component of an enzyme is called:

- A) Apoenzyme
- B) Coenzyme ✓
- C) Inhibitor
- D) Activator

Answer: B

4. The complete active enzyme with its cofactor is called:

- A) Apoenzyme
- B) Proenzyme
- C) Holoenzyme ✓
- D) Isoenzyme

Answer: C

5. The specific region on an enzyme where substrate binds is called:

- A) Allosteric site
- B) Active site ✓
- C) Binding site
- D) Catalytic site

Answer: B

6. Enzymes speed up chemical reactions by lowering:

- A) Temperature
- B) Pressure
- C) Activation energy ✓
- D) pH

Answer: C

7. The "lock and key" model of enzyme action was proposed by:

- A) Michaelis
- B) Menten
- C) Emil Fischer ✓
- D) Koshland

Answer: C

8. Which model of enzyme action involves induced fit?

- A) Lock and key model
- B) Induced fit model ✓
- C) Fluid mosaic model
- D) Allosteric model

Answer: B

9. Enzymes are denatured by:

- A) Low temperature
- B) High temperature ✓
- C) Substrate concentration
- D) Enzyme concentration

Answer: B

10. The optimum pH for most human enzymes is:

- A) 2
- B) 4
- C) 7.4 ✓
- D) 10

Answer: C

11. Pepsin works best at which pH?

- A) 2 ✓
- B) 7
- C) 8
- D) 9

Answer: A

12. Trypsin works best at which pH?

- A) 2
- B) 4
- C) 8 ✓
- D) 10

Answer: C

13. The enzyme that breaks down starch into maltose is:

- A) Amylase ✓
- B) Lipase
- C) Protease
- D) Nuclease

Answer: A

14. The enzyme that breaks down proteins into amino acids is:

- A) Amylase
- B) Lipase
- C) Protease ✓
- D) Nuclease

Answer: C

15. The enzyme that breaks down lipids into fatty acids and glycerol is:

- A) Amylase
- B) Lipase ✓
- C) Protease
- D) Nuclease

Answer: B

16. The phenomenon where the enzyme's shape changes after substrate binding is called:

- A) Denaturation
- B) Induced fit ✓
- C) Inhibition
- D) Activation

Answer: B

17. Reversible inhibitors bind to the enzyme through:

- A) Covalent bonds
- B) Ionic/Hydrogen bonds ✓
- C) Disulfide bonds
- D) Peptide bonds

Answer: B

18. Non-competitive inhibitors bind to the enzyme at:

- A) Active site
- B) Allosteric site ✓
- C) Substrate site
- D) Product site

Answer: B

19. The enzyme that is first used in DNA replication to unwind the DNA helix is:

- A) DNA polymerase
- B) Helicase ✓
- C) Ligase
- D) Primase

Answer: B

20. Enzymes are sensitive to changes in:

- A) Temperature only
- B) pH only
- C) Temperature and pH ✓
- D) None of these

Answer: C

BIOENERGETICS - 20 MCQs

1. The process of converting light energy into chemical energy is called:

- A) Respiration
- B) Photosynthesis ✓
- C) Fermentation
- D) Transpiration

Answer: B

2. The green pigment present in chloroplasts is called:

- A) Carotene
- B) Xanthophyll
- C) Chlorophyll ✓
- D) Phycobilin

Answer: C

3. The light-dependent reactions of photosynthesis occur in:

- A) Stroma
- B) Thylakoid membranes ✓
- C) Matrix
- D) Cytoplasm

Answer: B

4. The light-independent reactions (Calvin cycle) occur in:

- A) Thylakoid membranes

- B) Stroma ✓
- C) Cristae
- D) Nucleus

Answer: B

5. The source of oxygen released during photosynthesis is:

- A) Carbon dioxide
- B) Water ✓
- C) Glucose
- D) Oxygen

Answer: B

6. The end product of glycolysis is:

- A) Glucose
- B) Pyruvate ✓
- C) Acetyl CoA
- D) Citrate

Answer: B

7. The Krebs cycle occurs in:

- A) Cytoplasm
- B) Thylakoid
- C) Mitochondrial matrix ✓
- D) Nucleus

Answer: C

8. The electron transport chain is located in:

- A) Outer mitochondrial membrane
- B) Inner mitochondrial membrane ✓
- C) Stroma
- D) Cytoplasm

Answer: B

9. ATP is produced in the electron transport chain through:

- A) Substrate-level phosphorylation
- B) Oxidative phosphorylation ✓
- C) Photophosphorylation
- D) Glycolysis

Answer: B

10. The final electron acceptor in aerobic respiration is:

- A) NADH
- B) FADH₂
- C) Oxygen ✓
- D) Water

Answer: C

11. How many ATP molecules are produced from one glucose molecule during aerobic respiration?

- A) 2
- B) 8
- C) 36-38 ✓
- D) 40

Answer: C

12. The process of ATP production in chloroplasts is called:

- A) Oxidative phosphorylation
- B) Photophosphorylation ✓
- C) Substrate-level phosphorylation

D) Chemiosmosis

Answer: B

13. Which of the following is NOT a product of the light-dependent reactions?

- A) ATP
- B) NADPH
- C) Oxygen
- D) Glucose ✓

Answer: D

14. The enzyme that fixes CO₂ in the Calvin cycle is:

- A) ATP synthase
- B) Rubisco ✓
- C) Dehydrogenase
- D) Kinase

Answer: B

15. Anaerobic respiration occurs in the absence of:

- A) Glucose
- B) Oxygen ✓
- C) Water
- D) CO₂

Answer: B

16. The end product of anaerobic respiration in yeast is:

- A) Lactic acid
- B) Ethanol and CO₂ ✓
- C) Water

D) Glucose

Answer: B

17. The end product of anaerobic respiration in human muscles is:

- A) Ethanol
- B) Lactic acid ✓
- C) CO₂
- D) Pyruvate

Answer: B

18. The number of carbon atoms in pyruvate is:

- A) 2
- B) 3 ✓
- C) 4
- D) 6

Answer: B

19. FADH₂ is produced during which stage?

- A) Glycolysis
- B) Krebs cycle ✓
- C) Electron transport chain
- D) Calvin cycle

Answer: B

20. The main function of ATP is:

- A) Store genetic information
- B) Energy currency of the cell ✓
- C) Synthesize proteins
- D) Store carbohydrates

Answer: B

CELL CYCLE & CELL DIVISION - 20 MCQs

1. The cell cycle consists of:
- A) Interphase and M-phase ✓
 - B) Prophase and Metaphase
 - C) Anaphase and Telophase
 - D) S-phase and G-phase

Answer: A

2. During interphase, DNA replication occurs in:
- A) G₁ phase
 - B) S phase ✓
 - C) G₂ phase
 - D) M phase

Answer: B

3. The longest phase of the cell cycle is:
- A) M-phase
 - B) Interphase ✓
 - C) Prophase
 - D) Metaphase

Answer: B

4. Mitosis results in the formation of:
- A) Two haploid cells

- B) Two diploid cells ✓
- C) Four haploid cells
- D) Four diploid cells

Answer: B

5. Meiosis results in the formation of:
- A) Two haploid cells
 - B) Two diploid cells
 - C) Four haploid cells ✓
 - D) Four diploid cells

Answer: C

6. The phase where chromosomes align at the equator of the cell is:
- A) Prophase
 - B) Metaphase ✓
 - C) Anaphase
 - D) Telophase

Answer: B

7. The phase where sister chromatids separate and move to opposite poles is:
- A) Prophase
 - B) Metaphase
 - C) Anaphase ✓
 - D) Telophase

Answer: C

8. The nuclear envelope dissolves during:
- A) Prophase ✓
 - B) Metaphase
 - C) Anaphase

D) Telophase

Answer: A

9. The nuclear envelope reappears during:

- A) Prophase
- B) Metaphase
- C) Anaphase
- D) Telophase ✓

Answer: D

10. The spindle fibers are formed from:

- A) Centrioles ✓
- B) Ribosomes
- C) Mitochondria
- D) Nucleus

Answer: A

11. Crossing over occurs during which phase of meiosis?

- A) Prophase I ✓
- B) Metaphase I
- C) Anaphase I
- D) Telophase I

Answer: A

12. The reduction in chromosome number from diploid to haploid occurs during:

- A) Prophase I
- B) Metaphase I
- C) Anaphase I ✓
- D) Telophase I

Answer: C

13. Centromere divides during:

- A) Prophase I
- B) Metaphase I
- C) Anaphase II ✓
- D) Telophase II

Answer: C

14. The exchange of genetic material between homologous chromosomes is called:

- A) Translocation
- B) Crossing over ✓
- C) Mutation
- D) Duplication

Answer: B

15. Meiosis occurs in which type of cells?

- A) Somatic cells
- B) Germ cells ✓
- C) Epithelial cells
- D) Muscle cells

Answer: B

16. The number of chromosomes in human body cells is:

- A) 23
- B) 44
- C) 46 ✓
- D) 48

Answer: C

17. The number of chromosomes in human gametes is:

- A) 23 ✓
- B) 44
- C) 46
- D) 48

Answer: A

18. Cytokinesis in animal cells occurs by:

- A) Cell plate formation
- B) Cleavage furrow ✓
- C) Spindle formation
- D) Nucleus division

Answer: B

19. Cytokinesis in plant cells occurs by:

- A) Cleavage furrow
- B) Cell plate formation ✓
- C) Spindle formation
- D) Nuclear division

Answer: B

20. The spindle apparatus is formed during:

- A) Interphase
- B) Prophase ✓
- C) Metaphase
- D) Telophase

Answer: B

VARIETY OF LIFE - 20 MCQs

1. The branch of biology that deals with classification is called:

- A) Ecology
- B) Taxonomy ✓
- C) Genetics
- D) Embryology

Answer: B

2. The scientific name of an organism consists of:

- A) Genus and species ✓
- B) Family and genus
- C) Order and family
- D) Class and order

Answer: A

3. The system of classification based on evolutionary relationships is called:

- A) Artificial classification
- B) Natural classification
- C) Phylogenetic classification ✓
- D) Numerical classification

Answer: C

4. Two Kingdom System of classification was proposed by:

- A) Linnaeus ✓
- B) Whittaker
- C) Darwin

D) Lamarck

Answer: A

5. Five Kingdom System of classification was proposed by:

- A) Linnaeus
- B) Whittaker ✓
- C) Darwin
- D) Mendel

Answer: B

6. The Five Kingdoms according to Whittaker are:

- A) Monera, Protista, Fungi, Plantae, Animalia ✓
- B) Plantae, Animalia, Protista, Viruses, Monera
- C) Fungi, Monera, Protista, Algae, Animalia
- D) None of these

Answer: A

7. Viruses are placed in which kingdom?

- A) Monera
- B) Protista
- C) Fungi
- D) None (they are not placed in any kingdom) ✓

Answer: D

8. The largest group in biological classification is:

- A) Species

B) Genus

C) Kingdom ✓

D) Phylum

Answer: C

9. The smallest group in biological classification is:

- A) Species ✓
- B) Genus
- C) Kingdom
- D) Phylum

Answer: A

10. The correct sequence of taxonomic categories from broadest to narrowest is:

- A) Kingdom → Phylum → Class → Order → Family → Genus → Species ✓
- B) Species → Genus → Family → Order → Class → Phylum → Kingdom
- C) Kingdom → Class → Phylum → Order → Family → Genus → Species
- D) Kingdom → Phylum → Order → Class → Family → Genus → Species

Answer: A

11. The term "Binomial Nomenclature" was introduced by:

- A) Darwin
- B) Lamarck
- C) Linnaeus ✓
- D) Whittaker

Answer: C

12. In scientific name, the first word represents:

- A) Species
- B) Genus ✓
- C) Family
- D) Order

Answer: B

13. In scientific name, the second word represents:

- A) Species ✓
- B) Genus
- C) Family
- D) Order

Answer: A

14. The scientific name of human is:

- A) Homo sapiens ✓
- B) Homo erectus
- C) Homo habilis
- D) Homo neanderthalensis

Answer: A

15. The study of viruses is called:

- A) Virology ✓
- B) Bacteriology
- C) Mycology
- D) Parasitology

Answer: A

16. The study of bacteria is called:

- A) Virology
- B) Bacteriology ✓
- C) Mycology
- D) Parasitology

Answer: B

17. The study of fungi is called:

- A) Virology
- B) Bacteriology
- C) Mycology ✓
- D) Parasitology

Answer: C

18. The study of algae is called:

- A) Phycology ✓
- B) Mycology
- C) Bacteriology
- D) Virology

Answer: A

19. Viruses are considered as:

- A) Living
- B) Non-living
- C) Both living and non-living ✓
- D) None of these

Answer: C

20. The protein coat of a virus is called:

- A) Capsid ✓
- B) Capsomere
- C) Envelope
- D) Matrix

Answer: A

KINGDOM PROKARYOTAE (BACTERIA) - 20 MCQs

1. Bacteria are classified under which kingdom?

- A) Monera ✓
- B) Protista
- C) Fungi
- D) Plantae

Answer: A

2. Bacteria are:

- A) Eukaryotic
- B) Prokaryotic ✓
- C) Multicellular
- D) None of these

Answer: B

3. The genetic material in bacteria is:

- A) Linear DNA
- B) Circular DNA ✓
- C) RNA
- D) Both DNA and RNA

Answer: B

4. Bacteria reproduce by:

- A) Binary fission ✓
- B) Mitosis
- C) Meiosis
- D) Budding

Answer: A

5. The cell wall of bacteria is made up of:

- A) Cellulose
- B) Chitin
- C) Peptidoglycan ✓
- D) Protein

Answer: C

6. Gram-positive bacteria stain:

- A) Pink
- B) Purple ✓
- C) Green
- D) Yellow

Answer: B

7. Gram-negative bacteria stain:

- A) Pink/Red ✓
- B) Purple
- C) Green
- D) Yellow

Answer: A

8. The Gram staining technique was developed by:

- A) Robert Koch
- B) Louis Pasteur
- C) Christian Gram ✓

D) Joseph Lister

Answer: C

9. Bacteria that live in extreme environments are called:

- A) Eubacteria
- B) Archaeobacteria ✓
- C) Cyanobacteria
- D) Mycoplasma

Answer: B

10. Bacteria that are spiral-shaped are called:

- A) Cocci
- B) Bacilli
- C) Spirilla ✓
- D) Vibrio

Answer: C

11. Bacteria that are rod-shaped are called:

- A) Cocci
- B) Bacilli ✓
- C) Spirilla
- D) Vibrio

Answer: B

12. Bacteria that are spherical-shaped are called:

- A) Cocci ✓
- B) Bacilli
- C) Spirilla
- D) Vibrio

Answer: A

13. Bacteria that are comma-shaped are called:

- A) Cocci
- B) Bacilli
- C) Spirilla
- D) Vibrio ✓

Answer: D

14. The process of transfer of genetic material through conjugation involves:

- A) Pili ✓
- B) Flagella
- C) Capsule
- D) Cell wall

Answer: A

15. Bacteria which use sunlight for energy are called:

- A) Chemosynthetic
- B) Photosynthetic ✓
- C) Saprophytic
- D) Parasitic

Answer: B

16. Bacteria which obtain energy from chemical reactions are called:

- A) Chemosynthetic ✓
- B) Photosynthetic
- C) Saprophytic
- D) Parasitic

Answer: A

17. The outermost layer in some bacteria that provides protection is called:

- A) Cell wall
- B) Capsule ✓
- C) Plasma membrane
- D) Flagella

Answer: B

18. Bacteria that cause diseases are called:

- A) Pathogenic ✓
- B) Saprophytic
- C) Symbiotic
- D) Parasitic

Answer: A

19. The process of transformation in bacteria involves:

- A) Uptake of naked DNA ✓
- B) Transfer through pili
- C) Transfer through virus
- D) Binary fission

Answer: A

20. Bacteria that can survive without oxygen are called:

- A) Aerobic
- B) Anaerobic ✓
- C) Facultative
- D) None of these

Answer: B

KINGDOM ANIMALIA (BASICS) - 20 MCQs

1. Animals are:

- A) Autotrophic
- B) Heterotrophic ✓
- C) Photosynthetic
- D) None of these

Answer: B

2. The study of animals is called:

- A) Botany
- B) Zoology ✓
- C) Ecology
- D) Genetics

Answer: B

3. The basic unit of classification in animals is:

- A) Genus
- B) Species ✓
- C) Family
- D) Order

Answer: B

4. Animals without a backbone are called:

- A) Vertebrates

- B) Invertebrates ✓
- C) Chordates
- D) Mammals

Answer: B

5. Animals with a backbone are called:

- A) Vertebrates ✓
- B) Invertebrates
- C) Non-chordates
- D) None of these

Answer: A

6. The phylum Porifera includes:

- A) Jellyfish
- B) Sponges ✓
- C) Starfish
- D) Earthworm

Answer: B

7. The phylum Cnidaria includes:

- A) Sponges
- B) Jellyfish ✓
- C) Earthworm
- D) Snail

Answer: B

8. The phylum Platyhelminthes includes:

- A) Roundworms
- B) Flatworms ✓
- C) Earthworm
- D) Leech

Answer: B

9. The phylum Nematoda includes:

- A) Flatworms
- B) Roundworms ✓
- C) Earthworm
- D) Leech

Answer: B

10. The phylum Annelida includes:

- A) Roundworms
- B) Flatworms
- C) Earthworm ✓
- D) Snail

Answer: C

11. The phylum Arthropoda includes:

- A) Sponges
- B) Insects ✓
- C) Earthworm
- D) Snail

Answer: B

12. The phylum Mollusca includes:

- A) Insects
- B) Snails ✓
- C) Earthworm
- D) Starfish

Answer: B

13. The phylum Echinodermata includes:

- A) Insects
- B) Snails
- C) Starfish ✓
- D) Earthworm

Answer: C

14. The phylum Chordata includes:

- A) Insects
- B) Snails
- C) Starfish
- D) Vertebrates ✓

Answer: D

15. The five classes of vertebrates are:

- A) Pisces, Amphibia, Reptilia, Aves, Mammalia ✓
- B) Pisces, Reptilia, Aves, Mammalia, Arthropoda
- C) Amphibia, Reptilia, Aves, Mammalia, Annelida
- D) None of these

Answer: A

16. Fish belong to which class?

- A) Amphibia
- B) Reptilia
- C) Aves
- D) Pisces ✓

Answer: D

17. Frogs belong to which class?

- A) Pisces
- B) Amphibia ✓
- C) Reptilia
- D) Aves

Answer: B

18. Snakes belong to which class?

- A) Pisces
- B) Amphibia
- C) Reptilia ✓
- D) Aves

Answer: C

19. Birds belong to which class?

- A) Pisces
- B) Amphibia
- C) Reptilia
- D) Aves ✓

Answer: D

20. Mammals are characterized by the presence of:

- A) Feathers
- B) Scales
- C) Hair and mammary glands ✓
- D) Gills

Answer: C

HUMAN PHYSIOLOGY (BASICS) - 20 MCQs

1. The study of functions of the human body is called:

- A) Anatomy
- B) Physiology ✓
- C) Pathology
- D) Pharmacology

Answer: B

2. The study of structure of the human body is called:

- A) Anatomy ✓
- B) Physiology
- C) Pathology
- D) Pharmacology

Answer: A

3. The basic structural and functional unit of the human body is:

- A) Tissue
- B) Organ
- C) Cell ✓
- D) System

Answer: C

4. The group of similar cells performing a specific function is called:

- A) Organ
- B) Tissue ✓
- C) System
- D) Organelle

Answer: B

5. The largest organ of the human body is:

- A) Liver
- B) Heart
- C) Skin ✓
- D) Brain

Answer: C

6. The normal body temperature of humans is:

- A) 36°C
- B) 37°C ✓
- C) 38°C
- D) 39°C

Answer: B

7. The normal heart rate of an adult human is:

- A) 60-100 beats per minute ✓
- B) 100-150 beats per minute
- C) 150-200 beats per minute
- D) 200-250 beats per minute

Answer: A

8. The normal blood pressure of an adult human is:

- A) 80/120
- B) 120/80 ✓
- C) 100/80
- D) 140/100

Answer: B

9. The human heart is located in which cavity?

- A) Abdominal cavity
- B) Thoracic cavity ✓
- C) Cranial cavity
- D) Pelvic cavity

Answer: B

10. The human lungs are located in which cavity?

- A) Abdominal cavity
- B) Thoracic cavity ✓
- C) Cranial cavity
- D) Pelvic cavity

Answer: B

11. The human brain is located in which cavity?

- A) Abdominal cavity
- B) Thoracic cavity
- C) Cranial cavity ✓
- D) Pelvic cavity

Answer: C

12. The human stomach is located in which cavity?

- A) Abdominal cavity ✓
- B) Thoracic cavity
- C) Cranial cavity
- D) Pelvic cavity

Answer: A

13. The heart is made up of which type of muscle?

- A) Skeletal muscle
- B) Smooth muscle
- C) Cardiac muscle ✓
- D) None of these

Answer: C

14. The blood vessels that carry blood away from the heart are called:

- A) Veins
- B) Arteries ✓
- C) Capillaries
- D) Lymphatic vessels

Answer: B

15. The blood vessels that carry blood towards the heart are called:

- A) Veins ✓
- B) Arteries
- C) Capillaries
- D) Lymphatic vessels

Answer: A

16. The smallest blood vessels are called:

- A) Veins
- B) Arteries
- C) Capillaries ✓
- D) Venules

Answer: C

17. The liquid part of blood is called:

- A) Plasma ✓
- B) Serum
- C) Lymph
- D) Water

Answer: A

18. Red blood cells are also known as:

- A) Leukocytes
- B) Erythrocytes ✓
- C) Thrombocytes
- D) Lymphocytes

Answer: B

19. White blood cells are also known as:

- A) Leukocytes ✓
- B) Erythrocytes
- C) Thrombocytes
- D) Lymphocytes

Answer: A

20. Platelets are also known as:

- A) Leukocytes
- B) Erythrocytes
- C) Thrombocytes ✓
- D) Lymphocytes

Answer: C

DIGESTIVE SYSTEM - 20 MCQs

1. The process of breaking down food into simpler substances is called:

- A) Respiration
- B) Digestion ✓
- C) Excretion
- D) Circulation

Answer: B

2. The process of taking food into the body is called:

- A) Ingestion ✓
- B) Digestion
- C) Absorption
- D) Egestion

Answer: A

3. The process of passing undigested food out of the body is called:

- A) Ingestion
- B) Digestion
- C) Absorption
- D) Egestion ✓

Answer: D

4. The process of absorption of nutrients occurs mainly in:

- A) Stomach
- B) Small intestine ✓
- C) Large intestine
- D) Esophagus

Answer: B

5. The process of reabsorption of water occurs mainly in:

- A) Stomach
- B) Small intestine
- C) Large intestine ✓
- D) Esophagus

Answer: C

6. The wave-like contractions that propel food through the digestive tract are called:

- A) Peristalsis ✓
- B) Digestion
- C) Absorption
- D) Secretion

Answer: A

7. The enzyme in saliva that breaks down starch is:

- A) Pepsin
- B) Trypsin
- C) Amylase ✓
- D) Lipase

Answer: C

8. The enzyme present in gastric juice that breaks down proteins is:

- A) Pepsin ✓
- B) Trypsin
- C) Amylase
- D) Lipase

Answer: A

9. The enzyme present in pancreatic juice that breaks down proteins is:

- A) Pepsin
- B) Trypsin ✓
- C) Amylase
- D) Lipase

Answer: B

10. The enzyme present in pancreatic juice that breaks down fats is:

- A) Pepsin
- B) Trypsin
- C) Amylase
- D) Lipase ✓

Answer: D

11. Bile is produced by:

- A) Gallbladder
- B) Liver ✓
- C) Pancreas
- D) Stomach

Answer: B

12. Bile is stored in:

- A) Gallbladder ✓
- B) Liver
- C) Pancreas
- D) Stomach

Answer: A

13. The function of bile is to:

- A) Digest proteins
- B) Digest carbohydrates
- C) Emulsify fats ✓
- D) Digest nucleic acids

Answer: C

14. The finger-like projections in the small intestine that increase surface area for absorption are called:

- A) Villi ✓
- B) Microvilli
- C) Cilia
- D) Flagella

Answer: A

15. The acidic environment in the stomach is due to:

- A) Pepsin
- B) Hydrochloric acid ✓
- C) Bile
- D) Mucus

Answer: B

16. The function of mucus in the stomach is to:

- A) Digestion
- B) Absorption
- C) Protection of stomach lining ✓
- D) Emulsification

Answer: C

17. The digestion of carbohydrates begins in which organ?

- A) Mouth ✓
- B) Stomach
- C) Small intestine
- D) Large intestine

Answer: A

18. The digestion of proteins begins in which organ?

- A) Mouth
- B) Stomach ✓
- C) Small intestine
- D) Large intestine

Answer: B

19. The digestion of fats begins in which organ?

- A) Mouth
- B) Stomach
- C) Small intestine ✓
- D) Large intestine

Answer: C

20. The final absorption of nutrients occurs in:

- A) Stomach
- B) Small intestine ✓
- C) Large intestine
- D) Esophagus

Answer: B

CIRCULATORY SYSTEM - 20 MCQs

1. The circulatory system consists of:

- A) Heart, blood, blood vessels ✓
- B) Heart, lungs, brain
- C) Blood, lymph, liver
- D) Heart, kidneys, lungs

Answer: A

2. The human heart is:

- A) Two-chambered
- B) Three-chambered
- C) Four-chambered ✓
- D) Five-chambered

Answer: C

3. The chambers of the heart are:

- A) Two atria and two ventricles ✓
- B) One atrium and one ventricle
- C) Three atria and one ventricle
- D) One atrium and three ventricles

Answer: A

4. The upper chambers of the heart are called:

- A) Ventricles
- B) Atria ✓
- C) Valves
- D) Septum

Answer: B

5. The lower chambers of the heart are called:

- A) Ventricles ✓
- B) Atria
- C) Valves
- D) Septum

Answer: A

6. The blood vessels that carry blood away from the heart are called:

- A) Veins
- B) Arteries ✓
- C) Capillaries
- D) Lymphatic vessels

Answer: B

7. The blood vessels that carry blood towards the heart are called:

- A) Veins ✓
- B) Arteries
- C) Capillaries
- D) Lymphatic vessels

Answer: A

8. The smallest blood vessels are called:

- A) Veins
- B) Arteries
- C) Capillaries ✓
- D) Venules

Answer: C

9. The heart is made up of which type of muscle?

- A) Skeletal muscle
- B) Smooth muscle
- C) Cardiac muscle ✓
- D) None of these

Answer: C

10. The liquid part of blood is called:

- A) Plasma ✓
- B) Serum
- C) Lymph
- D) Water

Answer: A

11. Red blood cells are also known as:

- A) Leukocytes
- B) Erythrocytes ✓
- C) Thrombocytes
- D) Lymphocytes

Answer: B

12. White blood cells are also known as:

- A) Leukocytes ✓
- B) Erythrocytes
- C) Thrombocytes
- D) Lymphocytes

Answer: A

13. Platelets are also known as:

- A) Leukocytes
- B) Erythrocytes
- C) Thrombocytes ✓
- D) Lymphocytes

Answer: C

14. The function of red blood cells is to:

- A) Fight infections
- B) Transport oxygen ✓
- C) Clot blood
- D) Produce antibodies

Answer: B

15. The function of white blood cells is to:

- A) Transport oxygen
- B) Fight infections ✓
- C) Clot blood
- D) Carry nutrients

Answer: B

16. The function of platelets is to:

- A) Transport oxygen
- B) Fight infections
- C) Clot blood ✓
- D) Produce antibodies

Answer: C

17. The blood vessels that carry oxygenated blood from the heart to the body are called:

- A) Pulmonary arteries
- B) Pulmonary veins

- C) Aorta ✓
D) Vena cava

Answer: C

18. The blood vessels that carry deoxygenated blood from the body to the heart are called:

- A) Pulmonary arteries
B) Pulmonary veins
C) Aorta
D) Vena cava ✓

Answer: D

19. The blood vessels that carry deoxygenated blood from the heart to the lungs are called:

- A) Pulmonary arteries ✓
B) Pulmonary veins
C) Aorta
D) Vena cava

Answer: A

20. The blood vessels that carry oxygenated blood from the lungs to the heart are called:

- A) Pulmonary arteries
B) Pulmonary veins ✓
C) Aorta
D) Vena cava

Answer: B

RESPIRATORY SYSTEM - 20 MCQs

1. The process of taking in oxygen and releasing carbon dioxide is called:

- A) Respiration ✓
B) Digestion
C) Excretion
D) Circulation

Answer: A

2. The main organ of the respiratory system is:

- A) Heart
B) Liver
C) Lungs ✓
D) Kidney

Answer: C

3. The tube that carries air from the nose to the lungs is called:

- A) Esophagus
B) Trachea ✓
C) Pharynx
D) Bronchus

Answer: B

4. The trachea divides into:

- A) Two bronchi ✓
B) Three bronchi
C) Four bronchi
D) Five bronchi

Answer: A

5. The small air sacs in the lungs where gas exchange occurs are called:

- A) Bronchi
- B) Bronchioles
- C) Alveoli ✓
- D) Capillaries

Answer: C

6. The muscle that separates the chest cavity from the abdominal cavity is called:

- A) Intercostal muscles
- B) Diaphragm ✓
- C) Pectoral muscles
- D) Abdominal muscles

Answer: B

7. During inhalation, the diaphragm:

- A) Contracts and moves downward ✓
- B) Relaxes and moves upward
- C) Remains unchanged
- D) None of these

Answer: A

8. During exhalation, the diaphragm:

- A) Contracts and moves downward
- B) Relaxes and moves upward ✓
- C) Remains unchanged
- D) None of these

Answer: B

9. The exchange of gases occurs in the:

- A) Trachea
- B) Bronchi
- C) Alveoli ✓
- D) Larynx

Answer: C

10. The voice box is called:

- A) Trachea
- B) Bronchus
- C) Larynx ✓
- D) Pharynx

Answer: C

11. The flap that covers the trachea during swallowing is called:

- A) Epiglottis ✓
- B) Glottis
- C) Larynx
- D) Pharynx

Answer: A

12. Oxygen is transported in the blood by:

- A) White blood cells
- B) Platelets
- C) Hemoglobin ✓
- D) Plasma

Answer: C

13. The percentage of oxygen in inhaled air is approximately:

- A) 16%
- B) 21% ✓
- C) 30%
- D) 40%

Answer: B

14. The percentage of carbon dioxide in exhaled air is approximately:

- A) 0.04%
- B) 1%
- C) 4% ✓
- D) 16%

Answer: C

15. The respiratory center is located in the:

- A) Heart
- B) Lungs
- C) Brain ✓
- D) Spinal cord

Answer: C

16. The branches of trachea inside the lungs are called:

- A) Alveoli
- B) Bronchioles ✓
- C) Capillaries
- D) Bronchi

Answer: B

17. The outer covering of the lungs is called:

- A) Pericardium
- B) Pleura ✓

- C) Peritoneum
- D) Meninges

Answer: B

18. Cellular respiration takes place in:

- A) Cytoplasm
- B) Nucleus
- C) Mitochondria ✓
- D) Ribosome

Answer: C

19. The gas produced during cellular respiration is:

- A) Oxygen
- B) Nitrogen
- C) Carbon dioxide ✓
- D) Hydrogen

Answer: C

20. The gas consumed during cellular respiration is:

- A) Oxygen ✓
- B) Nitrogen
- C) Carbon dioxide
- D) Hydrogen

Answer: A

EXCRETORY SYSTEM - 20 MCQs

1. The process of removing metabolic waste products from the body is called:

- A) Digestion
- B) Respiration
- C) Excretion ✓
- D) Circulation

Answer: C

2. The main excretory organ of the human body is:

- A) Liver
- B) Kidney ✓
- C) Heart
- D) Lung

Answer: B

3. The functional unit of the kidney is called:

- A) Nephron ✓
- B) Neuron
- C) Alveolus
- D) Villus

Answer: A

4. The number of kidneys in the human body is:

- A) 1
- B) 2 ✓
- C) 3
- D) 4

Answer: B

5. The process of filtering blood in the kidney occurs in:

- A) Glomerulus ✓
- B) Collecting duct
- C) Loop of Henle
- D) Proximal tubule

Answer: A

6. The tube that carries urine from the kidney to the bladder is called:

- A) Urethra
- B) Ureter ✓
- C) Vas deferens
- D) Fallopian tube

Answer: B

7. The tube that carries urine from the bladder to the outside is called:

- A) Urethra ✓
- B) Ureter
- C) Vas deferens
- D) Fallopian tube

Answer: A

8. The process of removal of nitrogenous waste is called:

- A) Secretion
- B) Excretion ✓
- C) Absorption
- D) Digestion

Answer: B

9. The main nitrogenous waste in humans is:

- A) Ammonia
- B) Urea ✓
- C) Uric acid
- D) Creatinine

Answer: B

10. The liquid part of urine is:

- A) Blood
- B) Plasma
- C) Water ✓
- D) Lymph

Answer: C

11. The Bowman's capsule is part of the:

- A) Nephron ✓
- B) Alveoli
- C) Villi
- D) Neuron

Answer: A

12. The blood vessels that supply blood to the kidney are called:

- A) Pulmonary artery
- B) Renal artery ✓
- C) Hepatic artery
- D) Aorta

Answer: B

13. The blood vessels that drain blood from the kidney are called:

- A) Pulmonary vein
- B) Renal vein ✓
- C) Hepatic vein
- D) Vena cava

Answer: B

14. The process of reabsorption in the kidney occurs mainly in:

- A) Glomerulus
- B) Proximal convoluted tubule ✓
- C) Collecting duct
- D) Bowman's capsule

Answer: B

15. The hormone that regulates water reabsorption in the kidney is:

- A) Insulin
- B) ADH (Antidiuretic hormone) ✓
- C) Glucagon
- D) Thyroxine

Answer: B

16. The skin excretes waste products through:

- A) Nails
- B) Hair
- C) Sweat ✓
- D) Blood

Answer: C

17. The lungs excrete:

- A) Urea

- B) Carbon dioxide ✓
- C) Water
- D) Uric acid

Answer: B

18. The liver excretes:

- A) Urea
- B) Bile ✓
- C) Uric acid
- D) Ammonia

Answer: B

19. The pH of urine is approximately:

- A) 2-3
- B) 4-5
- C) 6-7 ✓
- D) 8-9

Answer: C

20. A healthy adult produces approximately how much urine per day?

- A) 0.5 liters
- B) 1-1.5 liters ✓
- C) 3-4 liters
- D) 5-6 liters

Answer: B

NERVOUS SYSTEM - 20 MCQs

1. The basic functional unit of the nervous system is:

- A) Nephron
- B) Neuron ✓
- C) Alveolus
- D) Villus

Answer: B

2. The central nervous system consists of:

- A) Brain and spinal cord ✓
- B) Brain and nerves
- C) Spinal cord and nerves
- D) All of these

Answer: A

3. The peripheral nervous system consists of:

- A) Brain
- B) Spinal cord
- C) Nerves ✓
- D) All of these

Answer: C

4. The brain is divided into how many main parts?

- A) 2
- B) 3 ✓
- C) 4
- D) 5

Answer: B

5. The three main parts of the brain are:

- A) Cerebrum, cerebellum, medulla oblongata ✓
- B) Cerebrum, thalamus, hypothalamus
- C) Cerebellum, pons, medulla
- D) Cerebrum, pons, medulla

Answer: A

6. The largest part of the brain is:

- A) Cerebellum
- B) Medulla oblongata
- C) Cerebrum ✓
- D) Hypothalamus

Answer: C

7. The part of the brain responsible for balance and coordination is:

- A) Cerebrum
- B) Cerebellum ✓
- C) Medulla oblongata
- D) Hypothalamus

Answer: B

8. The part of the brain that controls vital functions like breathing and heart rate is:

- A) Cerebrum
- B) Cerebellum
- C) Medulla oblongata ✓
- D) Hypothalamus

Answer: C

9. The functional unit of the nervous system is the:

- A) Neuron ✓
- B) Nephron
- C) Axon
- D) Dendrite

Answer: A

10. The part of the neuron that receives signals is the:

- A) Axon
- B) Dendrite ✓
- C) Cell body
- D) Myelin sheath

Answer: B

11. The part of the neuron that transmits signals away from the cell body is the:

- A) Axon ✓
- B) Dendrite
- C) Cell body
- D) Myelin sheath

Answer: A

12. The junction between two neurons is called:

- A) Synapse ✓
- B) Axon
- C) Dendrite
- D) Myelin sheath

Answer: A

13. The fatty sheath around the axon is called:

- A) Myelin sheath ✓
- B) Dendrite
- C) Synapse
- D) Cell body

Answer: A

14. The autonomic nervous system controls:

- A) Voluntary actions
- B) Involuntary actions ✓
- C) Both voluntary and involuntary actions
- D) None of these

Answer: B

15. The somatic nervous system controls:

- A) Voluntary actions ✓
- B) Involuntary actions
- C) Both voluntary and involuntary actions
- D) None of these

Answer: A

16. The sympathetic nervous system is responsible for:

- A) Rest and digest
- B) Fight or flight ✓
- C) Sleep and wake
- D) All of these

Answer: B

17. The parasympathetic nervous system is responsible for:

- A) Fight or flight
- B) Rest and digest ✓
- C) Voluntary actions
- D) None of these

Answer: B

18. Neurotransmitters are released from:

- A) Dendrites
- B) Axon terminal ✓
- C) Cell body
- D) Myelin sheath

Answer: B

19. The correct sequence of a reflex arc is:

- A) Receptor → Sensory neuron → Spinal cord → Motor neuron → Effector ✓
- B) Effector → Motor neuron → Spinal cord → Sensory neuron → Receptor
- C) Receptor → Motor neuron → Spinal cord → Sensory neuron → Effector
- D) Sensory neuron → Receptor → Spinal cord → Motor neuron → Effector

Answer: A

20. The three types of neurons are:

- A) Sensory, motor, interneuron ✓
- B) Sensory, motor, effector
- C) Motor, interneuron, effector
- D) Sensory, interneuron, effector

Answer: A

ENDOCRINE SYSTEM - 20 MCQs

1. The system of glands that secrete hormones is called:

- A) Nervous system
- B) Endocrine system ✓
- C) Circulatory system
- D) Digestive system

Answer: B

2. Chemical messengers secreted by endocrine glands are called:

- A) Enzymes
- B) Hormones ✓
- C) Neurotransmitters
- D) Antibodies

Answer: B

3. The master gland of the endocrine system is the:

- A) Thyroid gland
- B) Adrenal gland
- C) Pituitary gland ✓
- D) Pancreas

Answer: C

4. The pituitary gland is located in the:

- A) Neck
- B) Chest
- C) Brain ✓
- D) Abdomen

Answer: C

5. The gland that regulates metabolism is the:

- A) Thyroid gland ✓
- B) Adrenal gland
- C) Pituitary gland
- D) Pancreas

Answer: A

6. The hormone produced by the thyroid gland is:

- A) Insulin
- B) Thyroxine ✓
- C) Adrenaline
- D) Estrogen

Answer: B

7. The gland that controls the body's response to stress is the:

- A) Thyroid gland
- B) Adrenal gland ✓
- C) Pituitary gland
- D) Pancreas

Answer: B

8. The hormone produced by the adrenal gland during stress is:

- A) Insulin
- B) Thyroxine
- C) Adrenaline ✓
- D) Estrogen

Answer: C

9. The gland that regulates blood sugar levels is the:

- A) Thyroid gland
- B) Adrenal gland
- C) Pituitary gland
- D) Pancreas ✓

Answer: D

10. The hormone produced by the pancreas that lowers blood sugar is:

- A) Glucagon
- B) Insulin ✓
- C) Adrenaline
- D) Thyroxine

Answer: B

11. The hormone produced by the pancreas that raises blood sugar is:

- A) Glucagon ✓
- B) Insulin
- C) Adrenaline
- D) Thyroxine

Answer: A

12. The hormone that controls growth is produced by the:

- A) Thyroid gland
- B) Pituitary gland ✓
- C) Adrenal gland
- D) Pancreas

Answer: B

13. The gonads produce which hormones?

- A) Insulin and glucagon
- B) Estrogen and testosterone ✓
- C) Thyroxine and adrenaline
- D) None of these

Answer: B

14. Diabetes mellitus is caused by a deficiency of:

- A) Glucagon
- B) Insulin ✓
- C) Thyroxine
- D) Adrenaline

Answer: B

15. Goiter is caused by a deficiency of:

- A) Insulin
- B) Thyroxine ✓
- C) Adrenaline
- D) Glucagon

Answer: B

16. The hormone that promotes milk production in females is:

- A) Estrogen
- B) Progesterone
- C) Prolactin ✓
- D) Oxytocin

Answer: C

17. The hormone that causes the release of milk from the mammary glands is:

- A) Prolactin
- B) Oxytocin ✓
- C) Estrogen
- D) Progesterone

Answer: B

18. The hormone responsible for male secondary sexual characteristics is:

- A) Estrogen
- B) Testosterone ✓
- C) Progesterone
- D) Prolactin

Answer: B

19. The hormone responsible for female secondary sexual characteristics is:

- A) Estrogen ✓
- B) Testosterone
- C) Prolactin
- D) Oxytocin

Answer: A

20. The pituitary gland is often called the:

- A) Master gland ✓
- B) Stress gland
- C) Energy gland
- D) Growth gland

Answer: A

REPRODUCTION (HUMAN) - 20 MCQs

1. The process by which new organisms are produced is called:

- A) Respiration
- B) Reproduction ✓
- C) Digestion
- D) Excretion

Answer: B

2. The male reproductive organ is the:

- A) Ovary
- B) Uterus
- C) Testis ✓
- D) Vagina

Answer: C

3. The female reproductive organ is the:

- A) Testis
- B) Ovary ✓
- C) Penis
- D) Scrotum

Answer: B

4. The male gamete is called:

- A) Ovum
- B) Sperm ✓
- C) Zygote
- D) Embryo

Answer: B

5. The female gamete is called:

- A) Sperm
- B) Ovum ✓
- C) Zygote
- D) Embryo

Answer: B

6. The fusion of male and female gametes is called:

- A) Fertilization ✓
- B) Reproduction
- C) Division
- D) Growth

Answer: A

7. The fertilized egg is called:

- A) Sperm
- B) Ovum
- C) Zygote ✓
- D) Embryo

Answer: C

8. The development of the embryo takes place in the:

- A) Ovary
- B) Fallopian tube
- C) Uterus ✓
- D) Vagina

Answer: C

9. The male hormone is:

- A) Estrogen
- B) Testosterone ✓
- C) Progesterone
- D) Prolactin

Answer: B

10. The female hormone is:

- A) Testosterone
- B) Estrogen ✓
- C) Prolactin
- D) Oxytocin

Answer: B

11. The menstrual cycle occurs in:

- A) Males
- B) Females ✓
- C) Both males and females
- D) None of these

Answer: B

12. The menstrual cycle lasts approximately:

- A) 14 days
- B) 28 days ✓
- C) 30 days
- D) 365 days

Answer: B

13. Ovulation occurs around day:

- A) 7
- B) 14 ✓
- C) 21
- D) 28

Answer: B

14. The menopause occurs around age:
- A) 30-35
 - B) 40-45
 - C) 45-55 ✓
 - D) 60-65

Answer: C

15. The embryo develops into a fetus at approximately:
- A) 4 weeks
 - B) 8 weeks ✓
 - C) 12 weeks
 - D) 16 weeks

Answer: B

16. The gestation period in humans is approximately:
- A) 7 months
 - B) 8 months
 - C) 9 months ✓
 - D) 10 months

Answer: C

17. The process of birth is called:
- A) Fertilization
 - B) Gestation

- C) Parturition ✓
- D) Ovulation

Answer: C

18. The hormone that maintains pregnancy is:
- A) Estrogen
 - B) Testosterone
 - C) Progesterone ✓
 - D) Prolactin

Answer: C

19. The sperm is produced in the:
- A) Prostate gland
 - B) Seminal vesicles
 - C) Testes ✓
 - D) Vas deferens

Answer: C

20. The ovum is released from the:
- A) Uterus
 - B) Fallopian tube
 - C) Ovary ✓
 - D) Vagina

Answer: C

GENETICS & INHERITANCE - 20 MCQs

1. The study of heredity and variation is called:

- A) Ecology
- B) Genetics ✓
- C) Evolution
- D) Taxonomy

Answer: B

2. The father of genetics is:

- A) Darwin
- B) Mendel ✓
- C) Lamarck
- D) Linnaeus

Answer: B

3. The genetic material is:

- A) Protein
- B) DNA ✓
- C) RNA
- D) Lipid

Answer: B

4. The functional unit of heredity is the:

- A) Chromosome
- B) Gene ✓
- C) Nucleotide
- D) Protein

Answer: B

5. The number of chromosomes in humans is:

- A) 23

B) 44

C) 46 ✓

D) 48

Answer: C

6. The number of chromosomes in human gametes is:

A) 23 ✓

B) 44

C) 46

D) 48

Answer: A

7. The genotype of an organism refers to its:

A) Physical appearance

B) Genetic makeup ✓

C) Environment

D) Both A and B

Answer: B

8. The phenotype of an organism refers to its:

A) Physical appearance ✓

B) Genetic makeup

C) DNA sequence

D) All of these

Answer: A

9. An allele is:

A) A gene

B) A chromosome

C) An alternative form of a gene ✓

D) A protein

Answer: C

10. A dominant allele is expressed in:

- A) Only heterozygous condition
- B) Only homozygous condition
- C) Both homozygous and heterozygous condition ✓
- D) None of these

Answer: C

11. A recessive allele is expressed in:

- A) Only heterozygous condition
- B) Only homozygous condition ✓
- C) Both homozygous and heterozygous condition
- D) None of these

Answer: B

12. The Law of Segregation was proposed by:

- A) Darwin
- B) Mendel ✓
- C) Watson
- D) Crick

Answer: B

13. The Law of Independent Assortment was proposed by:

- A) Darwin
- B) Mendel ✓
- C) Lamarck

D) Linnaeus

Answer: B

14. The cross between two organisms with different traits is called:

- A) Monohybrid cross
- B) Dihybrid cross
- C) Test cross
- D) All of these ✓

Answer: D

15. A monohybrid cross involves:

- A) One trait ✓
- B) Two traits
- C) Three traits
- D) Four traits

Answer: A

16. A dihybrid cross involves:

- A) One trait
- B) Two traits ✓
- C) Three traits
- D) Four traits

Answer: B

17. The phenotypic ratio in a monohybrid cross is:

- A) 1:2:1
- B) 3:1 ✓
- C) 9:3:3:1
- D) 1:1

Answer: B

18. The genotypic ratio in a monohybrid cross is:

- A) 1:2:1 ✓
- B) 3:1
- C) 9:3:3:1
- D) 1:1

Answer: A

19. The phenotypic ratio in a dihybrid cross is:

- A) 1:2:1
- B) 3:1
- C) 9:3:3:1 ✓
- D) 1:1

Answer: C

20. Sex-linked traits are found on:

- A) Autosomes
- B) Sex chromosomes ✓
- C) Both autosomes and sex chromosomes
- D) None of these

Answer: B

MOLECULAR BIOLOGY (DNA/RNA) - 20 MCQs

1. DNA stands for:

- A) Deoxyribonucleic acid ✓
- B) Deoxyribose nucleic acid
- C) Deoxyribonucleotide acid
- D) None of these

Answer: A

2. RNA stands for:

- A) Ribonucleic acid ✓
- B) Ribose nucleic acid
- C) Ribonucleotide acid
- D) None of these

Answer: A

3. The shape of DNA is:

- A) Double helix ✓
- B) Single helix
- C) Triple helix
- D) None of these

Answer: A

4. The double helix model of DNA was proposed by:

- A) Mendel
- B) Darwin
- C) Watson and Crick ✓
- D) Lamarck

Answer: C

5. The monomers of DNA are:

- A) Amino acids
- B) Nucleotides ✓
- C) Fatty acids

D) Monosaccharides

Answer: B

6. The sugar found in DNA is:

- A) Ribose
- B) Deoxyribose ✓
- C) Glucose
- D) Sucrose

Answer: B

7. The sugar found in RNA is:

- A) Ribose ✓
- B) Deoxyribose
- C) Glucose
- D) Sucrose

Answer: A

8. The bases found in DNA are:

- A) Adenine, Thymine, Cytosine, Guanine ✓
- B) Adenine, Uracil, Cytosine, Guanine
- C) Adenine, Thymine, Uracil, Guanine
- D) None of these

Answer: A

9. The bases found in RNA are:

- A) Adenine, Thymine, Cytosine, Guanine
- B) Adenine, Uracil, Cytosine, Guanine ✓
- C) Adenine, Thymine, Uracil, Guanine
- D) None of these

Answer: B

10. The process of DNA copying itself is called:

- A) Transcription
- B) Translation
- C) Replication ✓
- D) Transformation

Answer: C

11. The process of making RNA from DNA is called:

- A) Replication
- B) Transcription ✓
- C) Translation
- D) Transformation

Answer: B

12. The process of making proteins from RNA is called:

- A) Replication
- B) Transcription
- C) Translation ✓
- D) Transformation

Answer: C

13. The enzyme that synthesizes DNA is:

- A) RNA polymerase
- B) DNA polymerase ✓
- C) Helicase
- D) Ligase

Answer: B

14. The enzyme that synthesizes RNA is:

- A) RNA polymerase ✓
- B) DNA polymerase
- C) Helicase
- D) Ligase

Answer: A

15. The enzyme that unwinds DNA during replication is:

- A) RNA polymerase
- B) DNA polymerase
- C) Helicase ✓
- D) Ligase

Answer: C

16. The codon is found on:

- A) DNA
- B) mRNA ✓
- C) tRNA
- D) rRNA

Answer: B

17. The anticodon is found on:

- A) DNA
- B) mRNA
- C) tRNA ✓
- D) rRNA

Answer: C

18. The genetic code is:

- A) Unambiguous

- B) Universal
- C) Degenerate
- D) All of these ✓

Answer: D

19. The start codon is:

- A) UAA
- B) UAG
- C) AUG ✓
- D) UGA

Answer: C

20. The stop codons are:

- A) UAA, UAG, UGA ✓
- B) AUG, UAA, UAG
- C) UGA, UAA, AUG
- D) None of these

Answer: A

BIOTECHNOLOGY & GENETIC ENGINEERING - 20 MCQs

1. The use of living organisms to produce useful products is called:

- A) Biotechnology ✓
- B) Genetic engineering
- C) Microbiology
- D) Ecology

Answer: A

2. The manipulation of genes is called:

- A) Biotechnology
- B) Genetic engineering ✓
- C) Microbiology
- D) Ecology

Answer: B

3. The enzyme used to cut DNA at specific sites is called:

- A) DNA polymerase
- B) Ligase
- C) Restriction enzyme ✓
- D) Helicase

Answer: C

4. The enzyme used to join DNA fragments is called:

- A) DNA polymerase
- B) Ligase ✓
- C) Restriction enzyme
- D) Helicase

Answer: B

5. A small circular DNA molecule found in bacteria is called:

- A) Chromosome
- B) Plasmid ✓
- C) Gene
- D) Nucleotide

Answer: B

6. The process of introducing foreign DNA into a host cell is called:

- A) Transformation ✓
- B) Transduction
- C) Conjugation
- D) Binary fission

Answer: A

7. The technique used to amplify DNA is called:

- A) Gel electrophoresis
- B) PCR (Polymerase Chain Reaction) ✓
- C) DNA sequencing
- D) Cloning

Answer: B

8. The process of separating DNA fragments by size is called:

- A) PCR
- B) Gel electrophoresis ✓
- C) DNA sequencing
- D) Cloning

Answer: B

9. A clone is:

- A) A genetically identical copy ✓
- B) A genetically different copy
- C) A mutated copy
- D) None of these

Answer: A

10. Dolly the sheep was produced by:

- A) PCR
- B) Genetic engineering
- C) Cloning ✓
- D) Transformation

Answer: C

11. Insulin is produced by genetic engineering in:

- A) Yeast
- B) Bacteria ✓
- C) Plants
- D) Animals

Answer: B

12. The process of making many copies of a gene is called:

- A) Gene cloning ✓
- B) Gene therapy
- C) Gene mutation
- D) Gene expression

Answer: A

13. The treatment of genetic diseases by replacing defective genes is called:

- A) Gene cloning
- B) Gene therapy ✓
- C) Gene mutation
- D) Gene expression

Answer: B

14. CRISPR is used for:

- A) DNA replication
- B) Gene editing ✓
- C) Protein synthesis
- D) Cell division

Answer: B

15. Transgenic organisms contain:

- A) Genes from the same species
- B) Genes from different species ✓
- C) No genes
- D) Only RNA

Answer: B

16. Bt cotton contains genes from:

- A) Bacteria ✓
- B) Virus
- C) Fungi
- D) Animals

Answer: A

17. PCR stands for:

- A) Polymerase Chain Reaction ✓
- B) Protein Chain Reaction
- C) Polymerase Chain Replication
- D) Protein Chain Replication

Answer: A

18. A vector is used to:

- A) Amplify DNA
- B) Carry foreign DNA into a host cell ✓
- C) Cut DNA
- D) Join DNA

Answer: B

19. The first genetically modified crop was:

- A) Maize
- B) Cotton
- C) Tomato ✓
- D) Soybean

Answer: C

20. Golden rice is genetically modified to contain:

- A) Protein
- B) Vitamin A ✓
- C) Vitamin C
- D) Iron

Answer: B

IMMUNITY - 20 MCQs

1. The ability of the body to fight against disease-causing organisms is called:

- A) Immunity ✓
- B) Digestion
- C) Respiration
- D) Excretion

Answer: A

2. The body's first line of defense includes:

- A) Skin and mucous membranes ✓
- B) White blood cells
- C) Antibodies
- D) Antigens

Answer: A

3. The second line of defense includes:

- A) Skin and mucous membranes
- B) Phagocytes and inflammation ✓
- C) Antibodies
- D) Lymphocytes

Answer: B

4. The third line of defense includes:

- A) Skin and mucous membranes
- B) Phagocytes and inflammation
- C) Antibodies and lymphocytes ✓
- D) All of these

Answer: C

5. White blood cells that engulf pathogens are called:

- A) Phagocytes ✓
- B) Lymphocytes
- C) Antibodies
- D) Platelets

Answer: A

6. Lymphocytes that produce antibodies are called:

- A) T-lymphocytes
- B) B-lymphocytes ✓

- C) Phagocytes
- D) Macrophages

Answer: B

7. Lymphocytes that kill infected cells are called:

- A) B-lymphocytes
- B) T-lymphocytes ✓
- C) Phagocytes
- D) Macrophages

Answer: B

8. Proteins produced by the body to fight antigens are called:

- A) Antigens
- B) Antibodies ✓
- C) Enzymes
- D) Hormones

Answer: B

9. A substance that triggers an immune response is called:

- A) Antigen ✓
- B) Antibody
- C) Enzyme
- D) Hormone

Answer: A

10. Immunity that is present at birth is called:

- A) Adaptive immunity
- B) Innate immunity ✓

- C) Acquired immunity
- D) Passive immunity

Answer: B

11. Immunity that develops after exposure to a pathogen is called:

- A) Innate immunity
- B) Adaptive immunity ✓
- C) Natural immunity
- D) Passive immunity

Answer: B

12. Active immunity involves:

- A) Antibodies from another source
- B) Body producing its own antibodies ✓
- C) Injection of antibodies
- D) None of these

Answer: B

13. Passive immunity involves:

- A) Body producing its own antibodies
- B) Antibodies from another source ✓
- C) Exposure to pathogen
- D) None of these

Answer: B

14. Vaccination provides:

- A) Passive immunity
- B) Active immunity ✓
- C) Innate immunity
- D) None of these

Answer: B

15. Vaccines contain:

- A) Live pathogens
- B) Weakened or killed pathogens ✓
- C) Antibodies
- D) Antigens

Answer: B

16. The immune system recognizes self from non-self by:

- A) MHC markers ✓
- B) Antibodies
- C) Antigens
- D) All of these

Answer: A

17. An autoimmune disease occurs when:

- A) Immune system attacks pathogens
- B) Immune system attacks the body's own cells ✓
- C) Immune system is weak
- D) None of these

Answer: B

18. HIV attacks:

- A) B-lymphocytes
- B) T-lymphocytes ✓
- C) Phagocytes
- D) Macrophages

Answer: B

19. AIDS is caused by:

- A) Bacteria
- B) Virus ✓
- C) Fungi
- D) Parasite

Answer: B

20. An allergic reaction is caused by:

- A) Overreaction of the immune system to harmless substances ✓
- B) Underreaction of the immune system
- C) Bacterial infection
- D) Viral infection

Answer: A